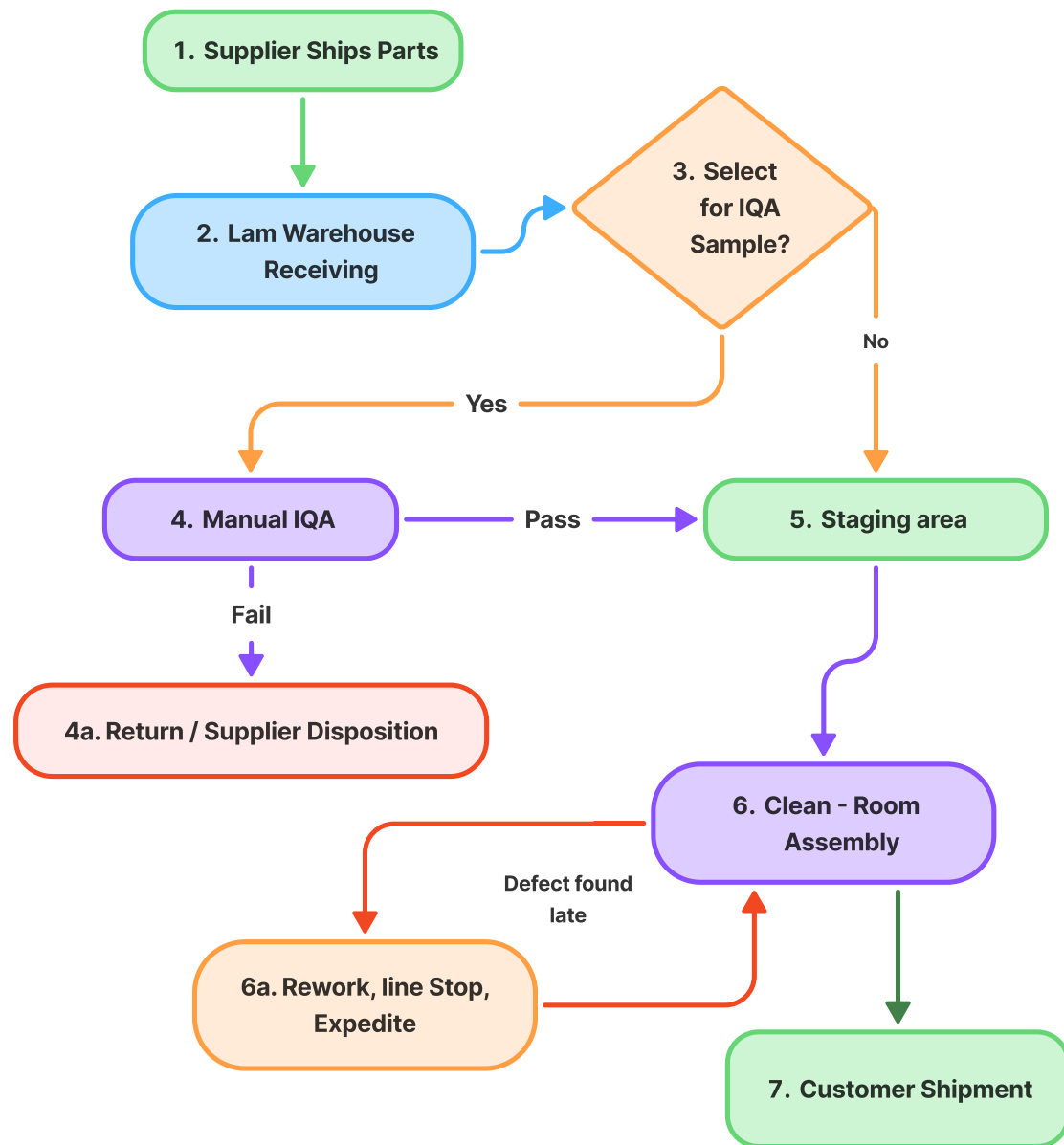
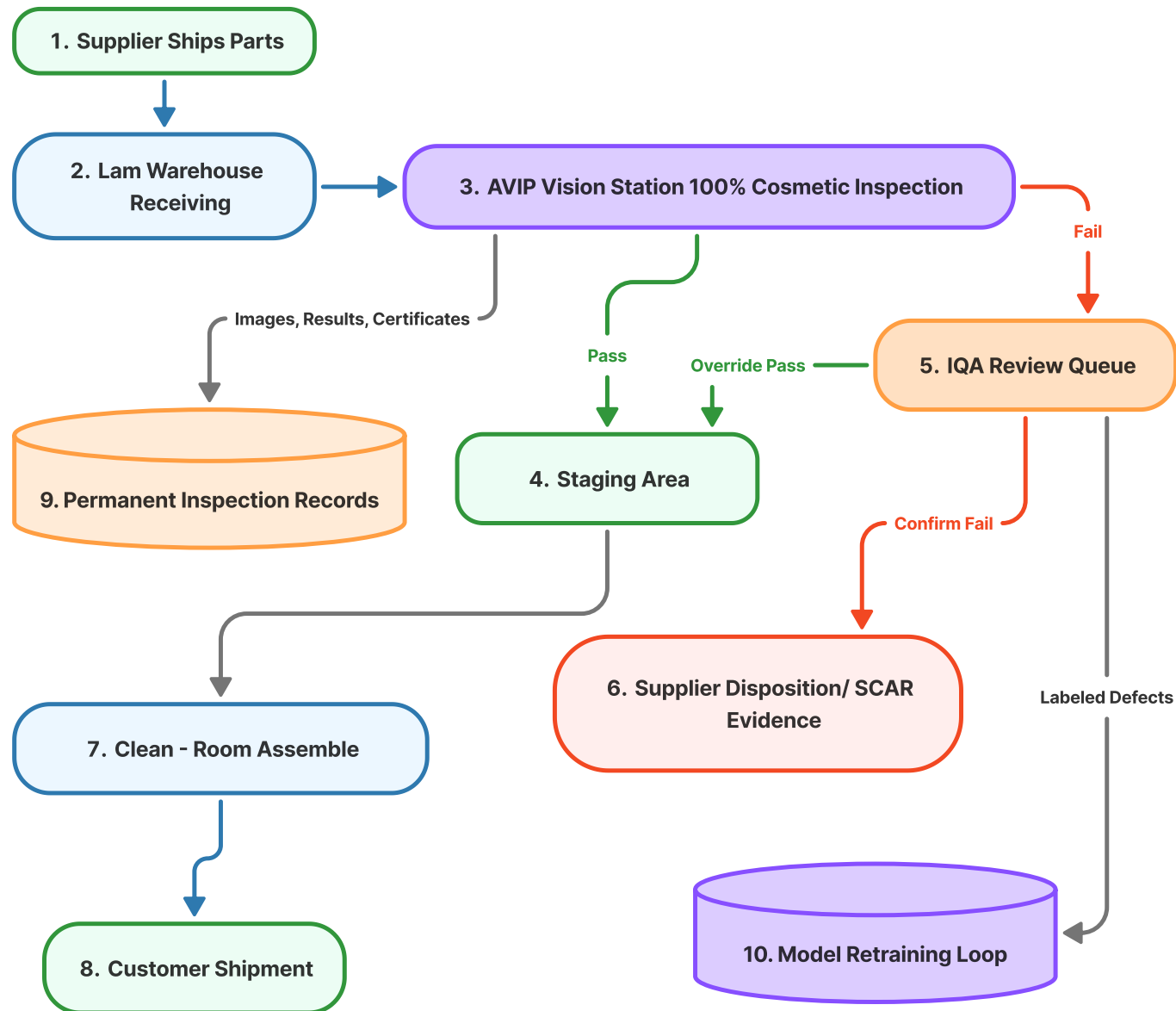


Lam RESEARCH - AI Vision Inspection Platform (AVIP)



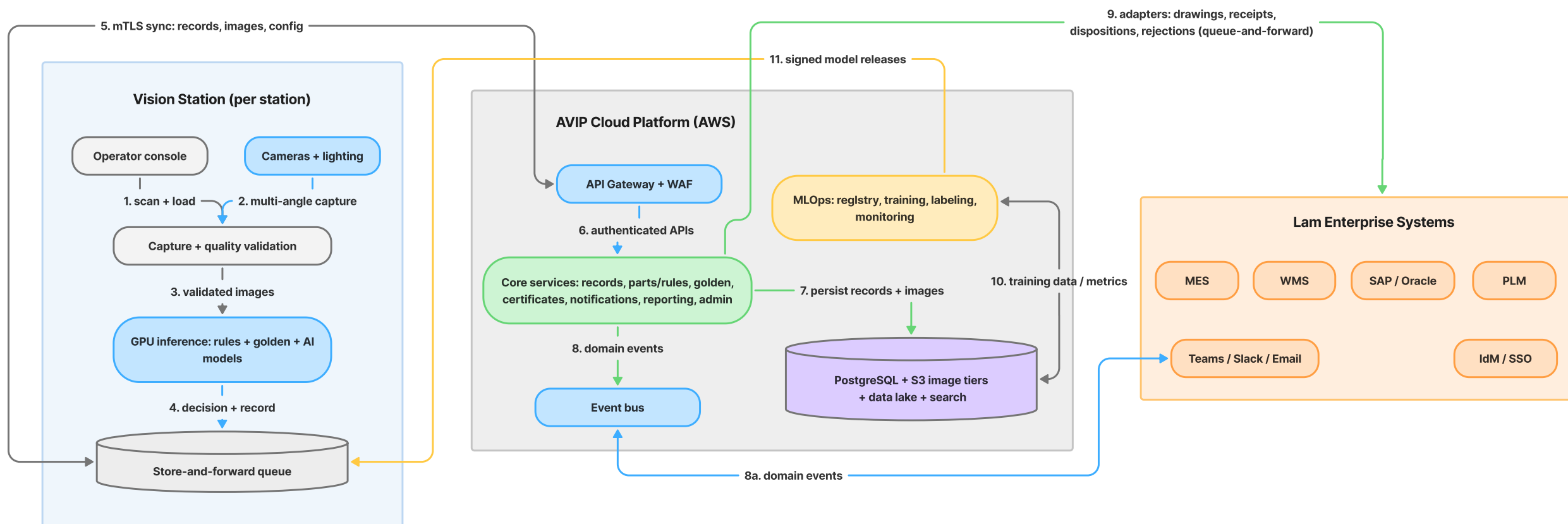
Step Description :

1. Supplier manufactures and ships parts to the assigned Lam warehouse.
2. Warehouse receives the shipment and books it against the purchase order.
3. A sampling rule decides whether this part is routed to IQA — only a minority of parts is selected.
4. Selected parts are manually inspected by IQA; passing parts continue to staging (5), failing parts go to supplier disposition (4a).
5. The majority of parts move directly from receiving to the staging area with no inspection at all.
6. Parts are kitted and consumed by clean-room assembly. Cosmetic defects are frequently discovered only here (6a), triggering rework, line stops, re-cleaning, and expedited replacements.
7. Completed systems ship to the customer.

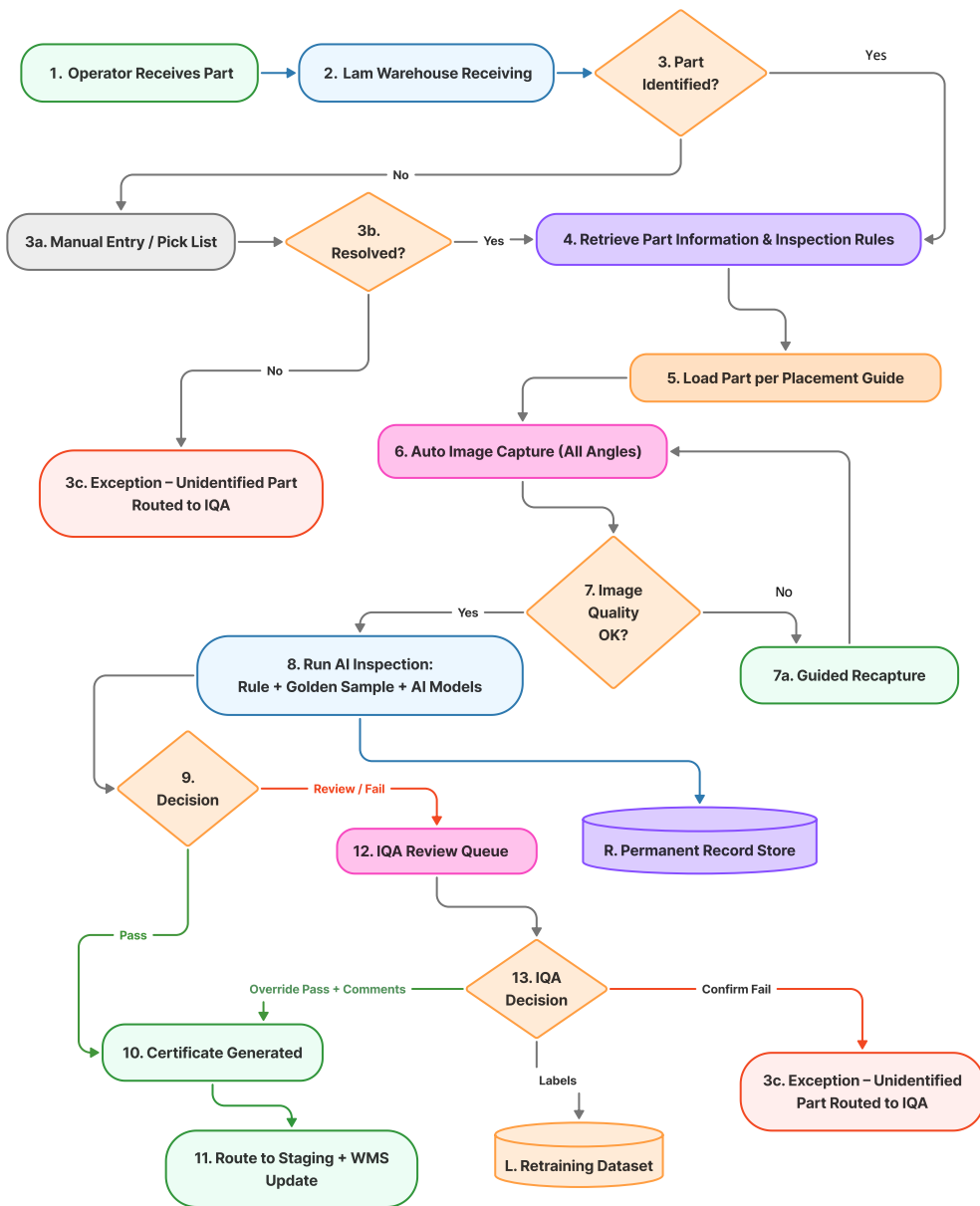


Step Description :

1. Supplier ships parts to the Lam warehouse (unchanged).
2. Warehouse receives the shipment against the purchase order (unchanged).
3. New: every part passes through an AVIP Vision Station for 100% cosmetic inspection before it can reach staging.
4. Parts that PASS route directly to staging with a signed inspection certificate.
5. Parts that FAIL enter the IQA review queue with all images, findings, and explanations attached.
6. IQA either overrides to PASS with a documented reason (part continues to staging) or confirms the FAIL, producing evidence-backed supplier disposition and SCAR input.
7. Assembly consumes only inspected parts — late discoveries drop sharply.
8. Completed systems ship to the customer.
9. Every inspection — images, results, certificates — is stored permanently for traceability.
10. Confirmed and overridden decisions become labeled data that continuously retrains and improves the AI models.

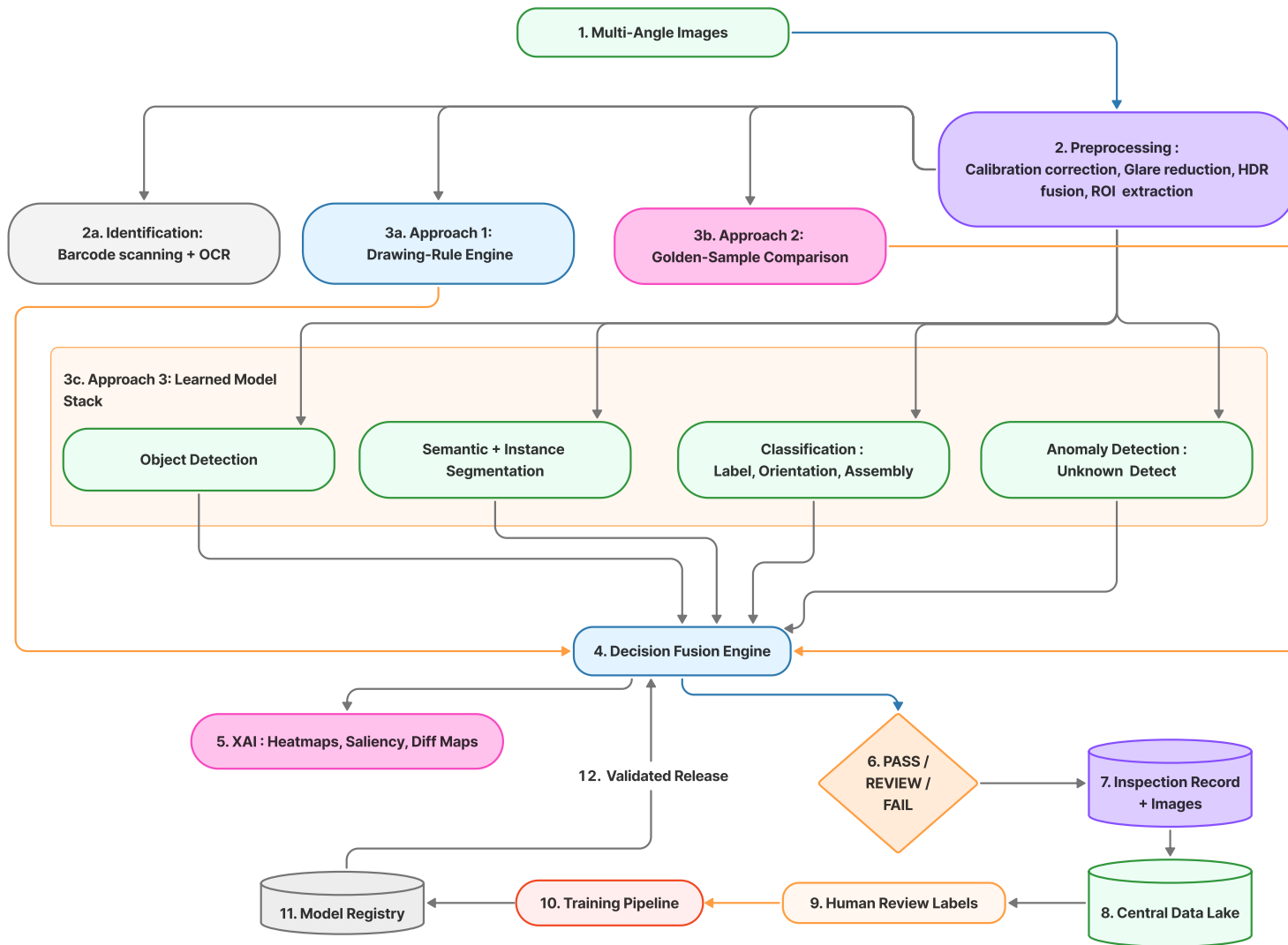


AVIP – AWS / Azure / GCP / Lam RESEARCH On-Premises



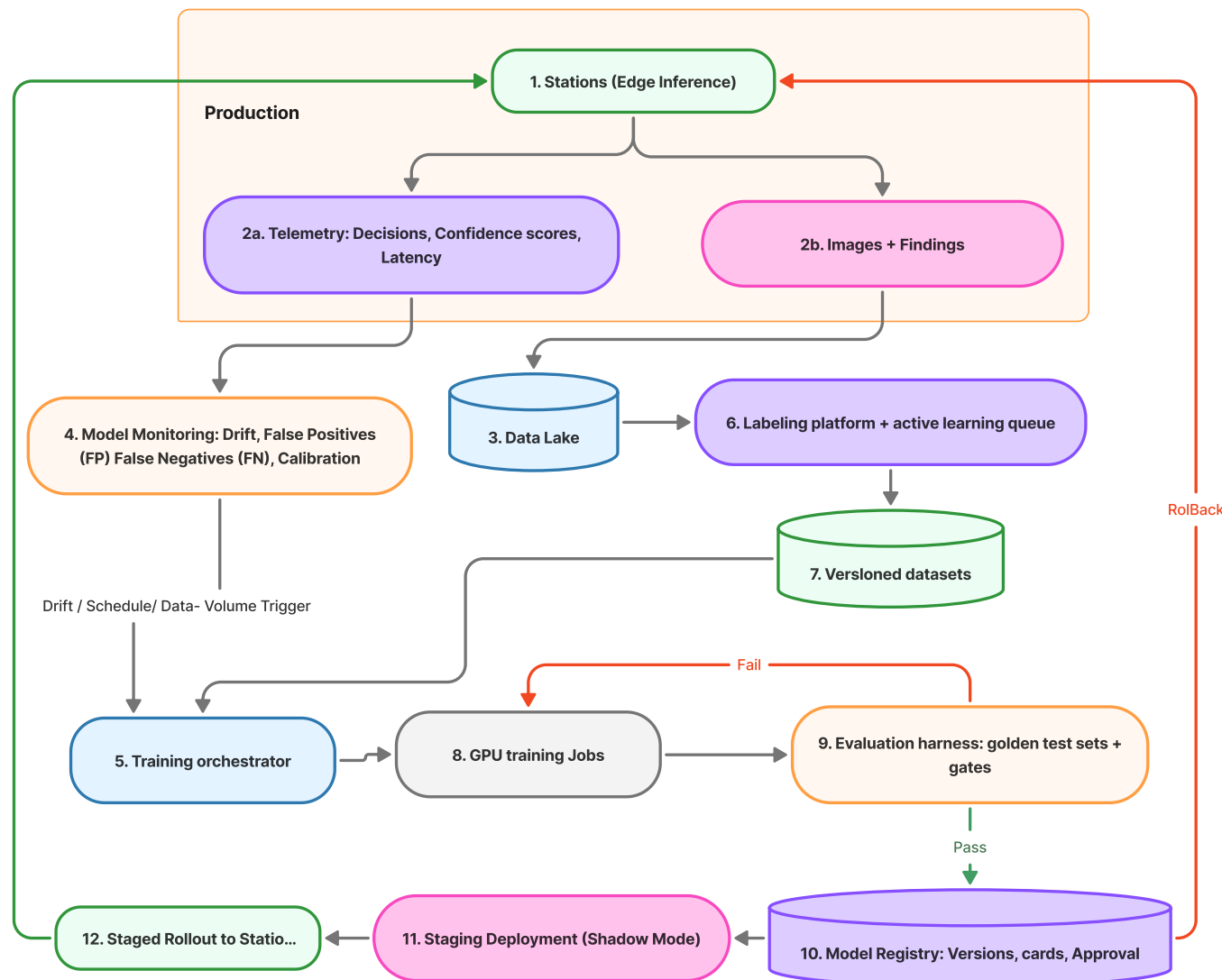
Step Description :

- Operator receives the part at the vision station.
- Operator scans the barcode/QR code; if the label is unreadable, OCR capture extracts candidate part numbers.
- The system resolves the identity. If not identified (3a), the operator uses manual entry with a validated pick list (3b); parts that still cannot be resolved become a logged exception and route to IQA (3c) — nothing bypasses inspection.
- The station retrieves the part's engineering data and published inspection rules for the exact drawing revision (from PLM or its local cache).
- Operator loads the part following the on-screen placement guide for that part family.
- All configured cameras fire automatically on part-presence detection; manual capture is also available.
- Each image passes automated quality validation (focus, exposure, glare, position). Failures trigger guided recapture (7a) that names the failing camera and check.
- The AI inspection executes on the station GPU in ≤ 10 seconds: drawing rules, golden-sample comparison, and learned models run together.
- The fusion engine produces one decision: PASS, REVIEW, or FAIL.
- For PASS (or an IQA override to PASS), a digitally signed inspection certificate is generated.
- The part routes to staging and the WMS disposition updates automatically.
- FAIL/REVIEW parts enter the prioritized IQA review queue with all evidence attached; a quarantine label is printed.
- The IQA inspector adjudicates: override to PASS with a mandatory reason code and comment, or confirm the FAIL.
- Confirmed failures move to quarantine/supplier disposition with evidence for supplier recovery.
- Every inspection — images, findings, decisions, model versions, calibration state — is stored permanently.
- Reviewer decisions are captured as labels that feed the model retraining dataset.



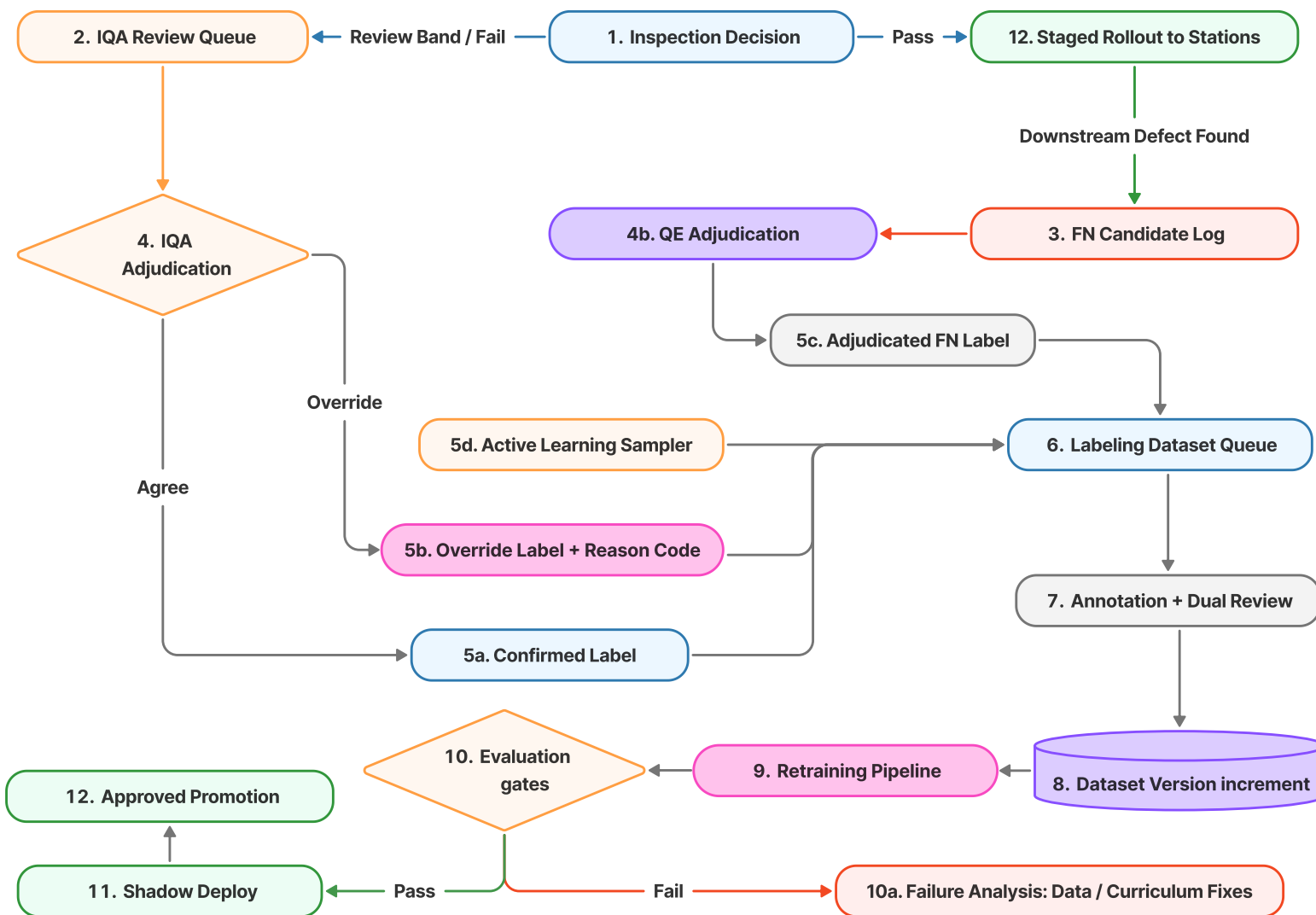
Step Description :

1. Multi-angle images arrive from the capture service after quality validation.
2. Preprocessing normalizes the images: calibration correction, glare reduction, HDR fusion, and region-of-interest extraction. Identification (2a) decodes barcodes and runs OCR in parallel.
3. The three inspection approaches execute concurrently: the drawing-rule engine (3a), golden-sample comparison (3b), and the learned model stack (3c) — object detection, segmentation, classification, and anomaly detection.
4. The decision fusion engine combines all findings under the per-family fusion rules (§3.5).
5. Explainability artifacts are generated for every finding: localization overlays, heatmaps, and golden-difference maps.
6. The fused result becomes the station decision: PASS, REVIEW, or FAIL.
7. The complete inspection record with images is written locally and queued for sync.
8. Records and images synchronize to the central data lake.
9. Human reviewers adjudicate REVIEW/FAIL cases, producing confirmed labels.
10. The training pipeline retrains models on the growing labeled dataset.
11. Passing models are versioned in the model registry with metrics and approvals.
12. Validated releases deploy back to stations — closing the continuous improvement loop.



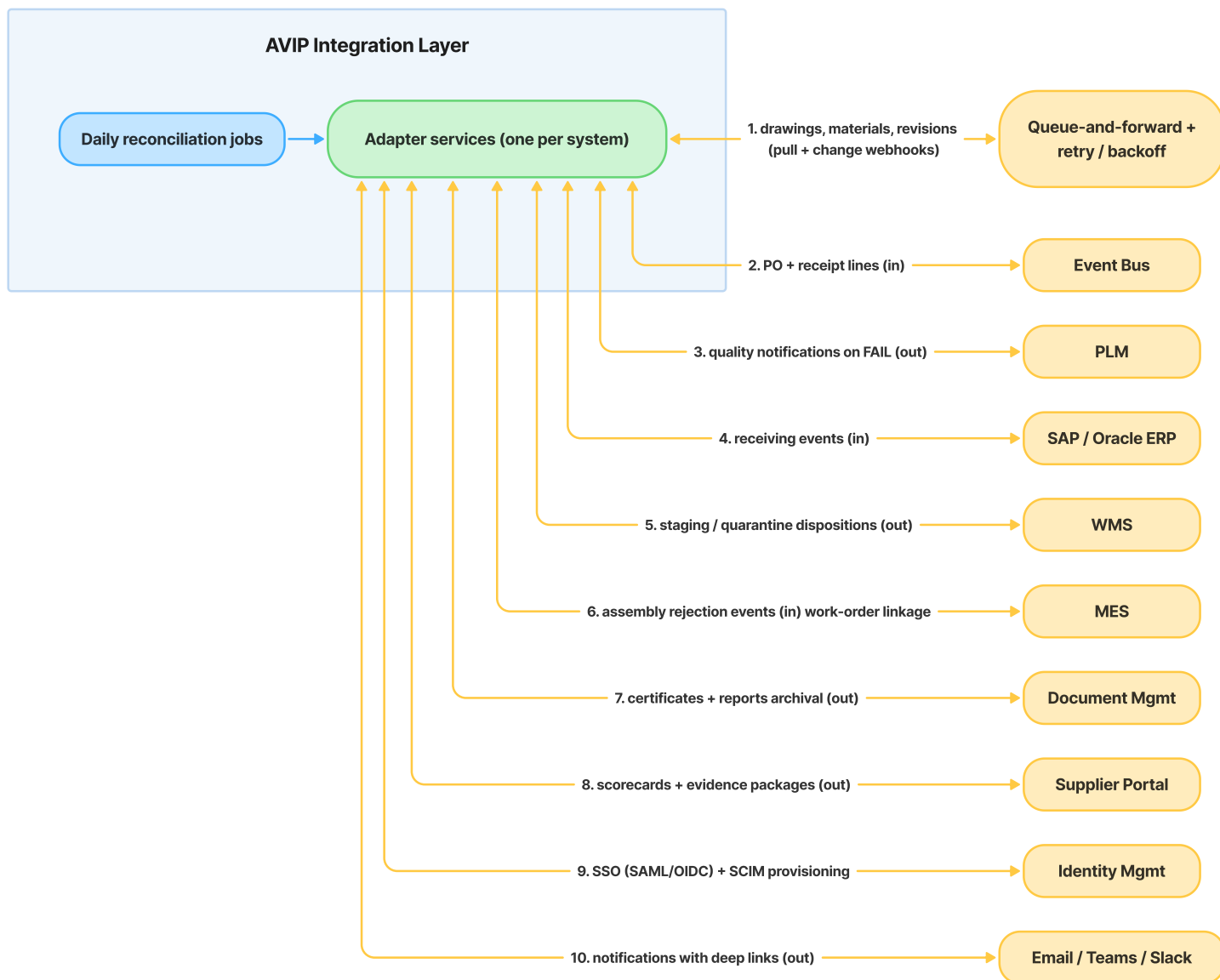
Step Description :

1. Production stations run edge inference on every part.
2. Each inspection emits telemetry — decisions, confidence, latency (2a) — and the images with findings (2b).
3. Images and findings land in the central data lake.
4. Model monitoring watches for input drift, accuracy proxies, FP/FN trends, and calibration decay.
5. The training orchestrator starts a retraining cycle when triggered by drift, schedule, or labeled-data volume.
6. The labeling platform, prioritized by active learning, turns production images into annotations.
7. Annotations are frozen into immutable, versioned datasets.
8. GPU training jobs retrain the affected models.
9. The evaluation harness scores candidates against frozen golden test sets and the release gates (§5.3); failures return to the orchestrator with analysis.
10. Passing models are registered with model cards, metrics, and dual approval (AI engineering + Quality).
11. New versions deploy first in shadow mode — scoring live traffic without affecting decisions.
12. After shadow validation and approvals, a staged rollout reaches the fleet, with automatic halt and rollback on regression.



Step Description :

- Every inspection ends in a decision: PASS, REVIEW, or FAIL.
- REVIEW and FAIL cases enter the IQA queue; PASS parts continue to staging (2b).
- If a passed part is later found defective downstream (IQA sampling, assembly rejection via MES), it is logged as a false-negative candidate.
- IQA adjudicates queued cases (4); Quality Engineering adjudicates FN candidates weekly (4b).
- Adjudication produces labeled ground truth: confirmed labels (5a), override labels with reason codes (5b), adjudicated FN labels (5c), plus active-learning selections from routine production images (5d).
- All labels flow into the prioritized labeling dataset queue.
- Annotators label with dual review; golden-set items require QE adjudication and $\kappa \geq 0.85$ agreement.
- The dataset version increments immutably — full lineage preserved.
- The retraining pipeline consumes the new dataset version.
- Candidates face the evaluation gates; failures go to root-cause analysis and data fixes (10a).
- Passing models run in shadow against live traffic.
- With shadow results clean and dual approval, the model is promoted — measurably better than its predecessor.



Step Description :

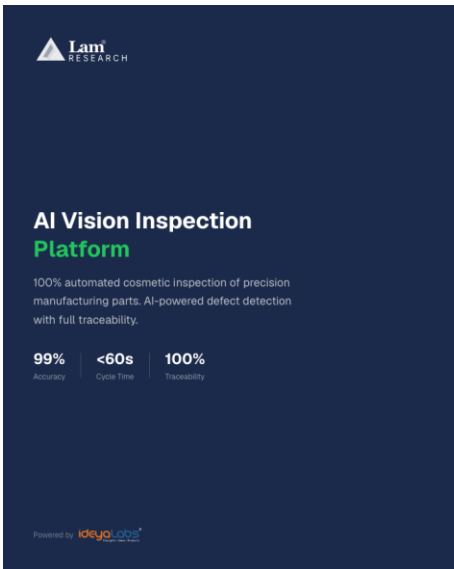
1. The PLM adapter pulls drawings, materials, and revision metadata by part number, and subscribes to revision-change webhooks that invalidate station caches and flag affected rules. A nightly delta sync catches anything missed.
2. The ERP adapter consumes purchase-order and receipt-line data so every scanned part is validated against what was actually ordered.
3. When IQA confirms a FAIL, the ERP adapter emits a quality notification into SAP/Oracle so supplier recovery starts in the existing quality process.
4. The WMS adapter consumes receiving events, pre-staging the inspection queue before the part even reaches the station.
5. On every decision, the WMS adapter writes the disposition back — staging for PASS, quarantine for FAIL — so warehouse inventory always reflects inspection status.
6. The MES adapter links inspection records to assembly work orders and consumes assembly rejection events, which feed the false-negative tracking loop (§5.6).
7. Signed certificates and reports archive automatically to Document Management with retention metadata.
8. Supplier scorecards and evidence packages publish to the Supplier Portal; in Phase 4 this channel also handles supplier station enrollment.
9. Identity Management provides single sign-on (SAML/OIDC) and automated user provisioning/deprovisioning (SCIM) with group-to-role mapping.
10. The notification service delivers event-driven alerts to Teams, Slack, and email, each deep-linking to the relevant record or dashboard.

AI Vision Inspection Platform (AVIP) – Fremont / Livermore Station

Hardware Recommendation

BOM Component (Qty)	Recommended Vendor & Product	Why Recommended	Alternative Vendors	Approx. Cost / Station
Area-scan cameras (×5)	Basler — ace 2 a2A4200-12gmPRO (12.2 MP GigE, global shutter)	Best balance of image quality, GenICam/pylon SDK maturity, lens ecosystem, and US support; standardizing one brand fleet-wide keeps capture profiles, calibration, and spares uniform	Teledyne FLIR (Blackfly S/Oryx), Allied Vision, IDS, Baumer	~\$5,500 (\$4.5K–7.5K)
Lenses (×5)	Edmund Optics — TECHSPEC HP series	Resolving power matched to 12 MP+ sensors, <0.1% distortion, US stock for fast iteration during station design; telecentric options for registration-critical families	Computar, Fujinon, Moritex (telecentric), Kowa	~\$3,000 (\$2K–5K)
Ring lighting (×2)	CCS Inc. — HPR2/LDR2 series	Industry's largest catalog and best uniformity; strobe/overdrive-ready; global support footprint matches the Phase-3/4 rollout map	Smart Vision Lights (integrated driver), Advanced Illumination, Effilux, TPL Vision	~\$2,000 (\$1K–3K)
Diffuse lighting (×1–2)	CCS Inc. — HPD2 dome / LFX3 flat dome	Reference-class "cloudy-day" uniformity for reflective parts; flat-dome variant permits camera-through-hole at longer working distances; single lighting vendor simplifies recipes	Advanced Illumination, Smart Vision Lights, Spectrum Illumination (oversized parts)	~\$1,800 (\$0.8K–4K)
Polarization kits	MidOpt — PR032 + light films	De-facto machine-vision filter standard, every thread size, durable coatings	Edmund Optics, CCS, Schneider-Kreuznach	~\$500 (\$0.3K–0.9K)
Lighting controller (×1)	Gardasoft — RT/TR-HT (4–8 ch)	Market-leading timing precision (to 1 µs), Ethernet API for capture-profile recipes, Trinitri integration with CCS lights	CCS controllers; Smart Vision Lights integrated drivers can eliminate this line entirely	~\$1,800 (\$1.2K–2.5K)
GPU workstation (×1)	Neosys — Nuvo-10108GC (RTX-class GPU, TPM 2.0)	Purpose-built rugged edge-AI platform: –25 to 60 °C, MIL-STD-810H, 350 W GPU budget headroom for the full model stack at ≤ 10 s/part	OnLogic (US configure-to-order), Advantech (global availability), Vecow, Dell Precision (IT-standard option)	~\$7,500 (\$4.5K–9.5K)
Touchscreen console (×1)	Hope Industrial — HIS-UM22 (22")	Fully enclosed industrial build, glove-capable PCAP, 5-year warranty — matched to warehouse-floor conditions	Advantech, Winmate, Teguar, Elo (needs enclosure integration)	~\$1,800 (\$1.4K–2.2K)
DPM barcode scanner (×1)	Zebra — DS3608-DP	Purpose-built for direct part marks (dot-peen, laser-etch), IP67, ubiquitous US supply and spares	Cognex DataMan 8700DX (strongest on hard DPM — fallback if pilot finds difficult marks), Keyence, Honeywell, Datalogic	~\$850 (\$0.7K–1K)
Enclosure / frame (×1)	Bosch Rexroth — aluminum framing + panels to ideyaLabs drawings	Global framing standard enables identical station rebuilds on three continents; ESD accessories; light-sealed matte interior per spec	80/20 Inc. (US, cost-effective), item (EU), welded fabrication (pilot), Terra Universal (Phase-5 clean-room variant)	~\$10,000 (\$6K–15K)
UPS (×1)	APC (Schneider) — Smart-UPS SMT2200	Managed shutdown integration, network card, global RMA; likely already a Lam IT standard	Eaton, Vertiv, CyberPower	~\$1,300 (\$0.8K–1.9K)
Label printer (×1)	Zebra — ZT411 industrial	Industrial duty cycle; Link-OS enables fleet-wide remote management of every station printer	Zebra ZD621 (lighter duty), SATO, Honeywell	~\$2,000 (\$0.6K–2.6K)
Industrial switch (×1)	Cisco — Catalyst IE3300	Aligns with likely Lam plant-network standards, inheriting existing support contracts; VLAN isolation for the camera LAN	Moxa, Hirschmann, Phoenix Contact	~\$2,000 (\$0.5K–3.5K)
Calibration kit (1/site)	Calib.io targets + Konica Minolta T-10A meter	Certified target flatness/accuracy for camera calibration; reference-class illuminance measurement for lighting calibration	Edmund Optics targets, Calibrite gray cards, Applied Image custom	~\$2,500 (\$1.5K–3.5K)
Approx. Total Cost Per Station				~\$47,000 (\$26K – \$71K)

AI Vision Inspection Platform (AVIP) – Application Walkthrough



Welcome back
Sign in to the AI Vision Inspection Platform

Email or Username

Password

Sign In

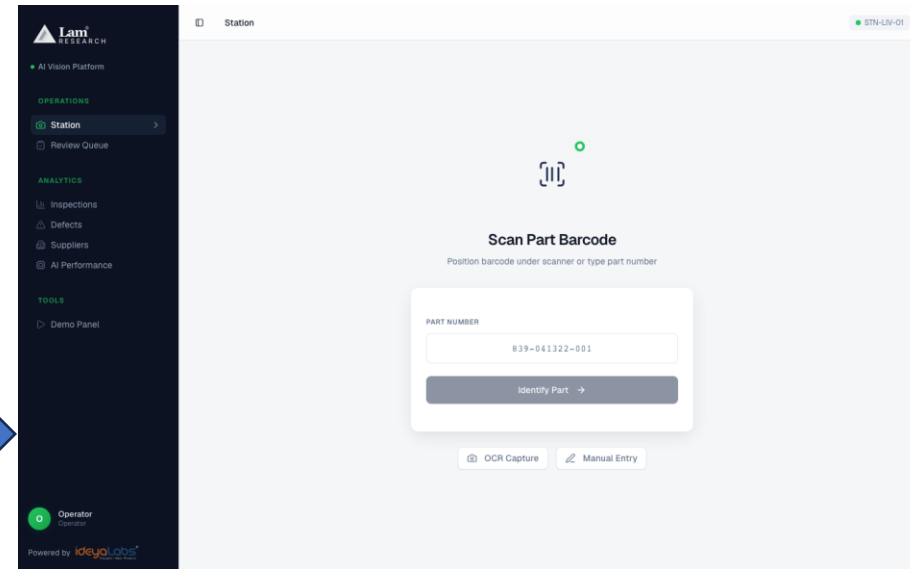
Production uses Lam SSO (SAML/OIDC)

DEMO QUICK ACCESS

Alex Chen Operator	Maria Santos QA Inspector
David Kim Quality Engineer	Sarah Johnson Admin

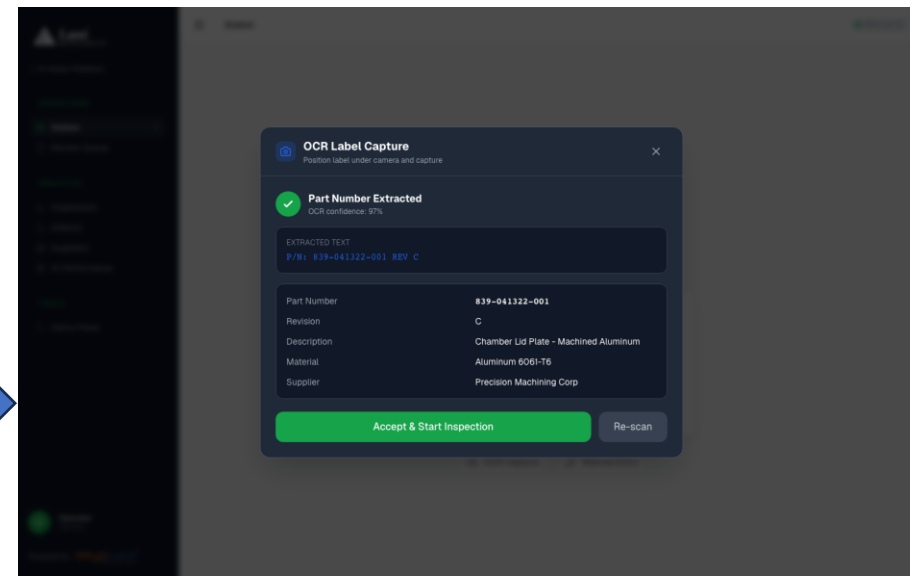
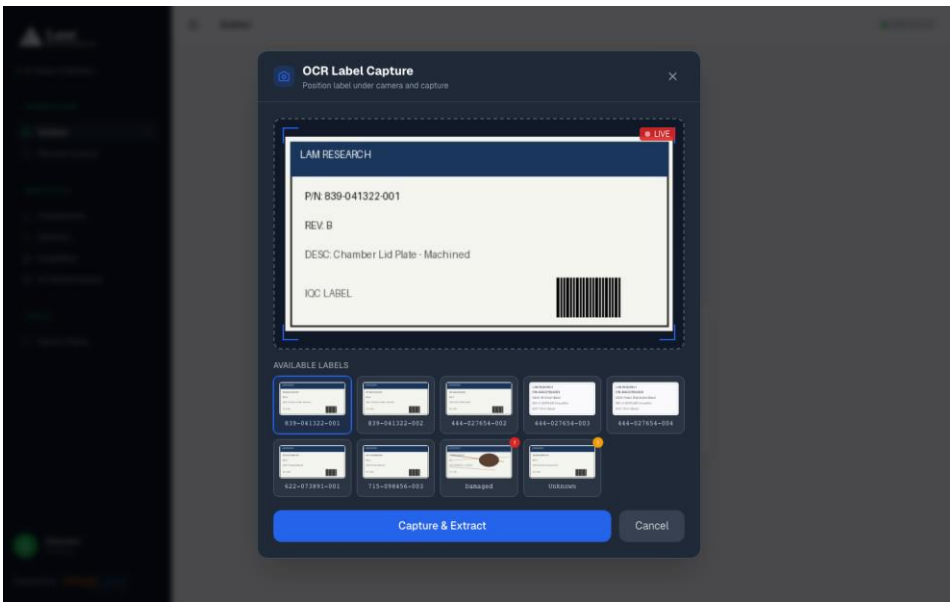
Step 1 – Login
🔒 Secure authentication for authorized operators..

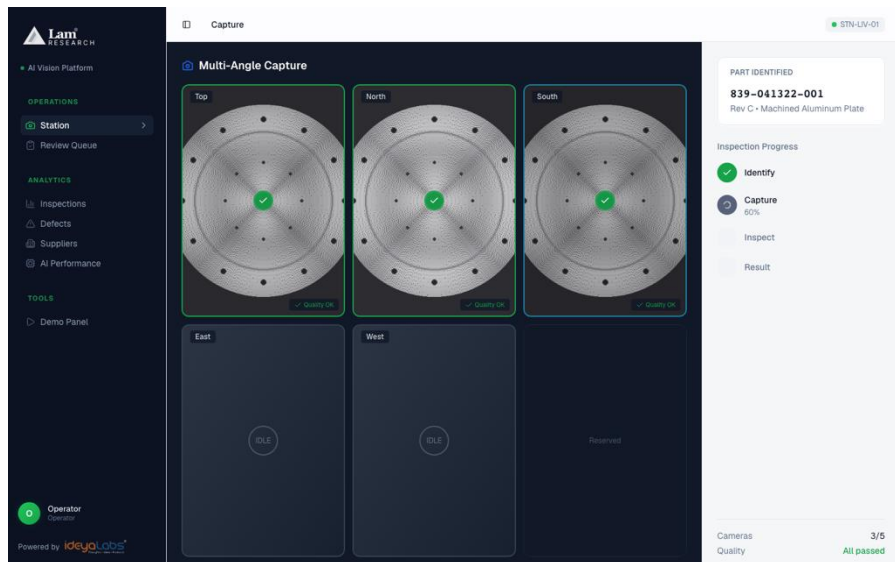
Step 2 – Identify Part
📦 Scan the barcode or part number to retrieve the correct inspection profile.



Step 3 – Validate Label
🏷️ OCR extracts and verifies labels, part numbers, and revision information.

Step 4 – OCR-Based Part Validation
Automatically extracts and validates part information before AI inspection.



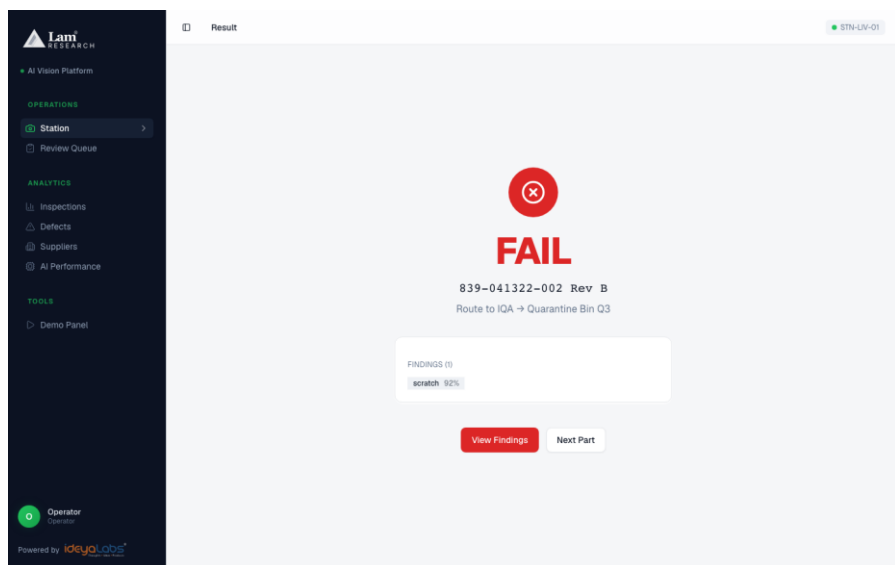
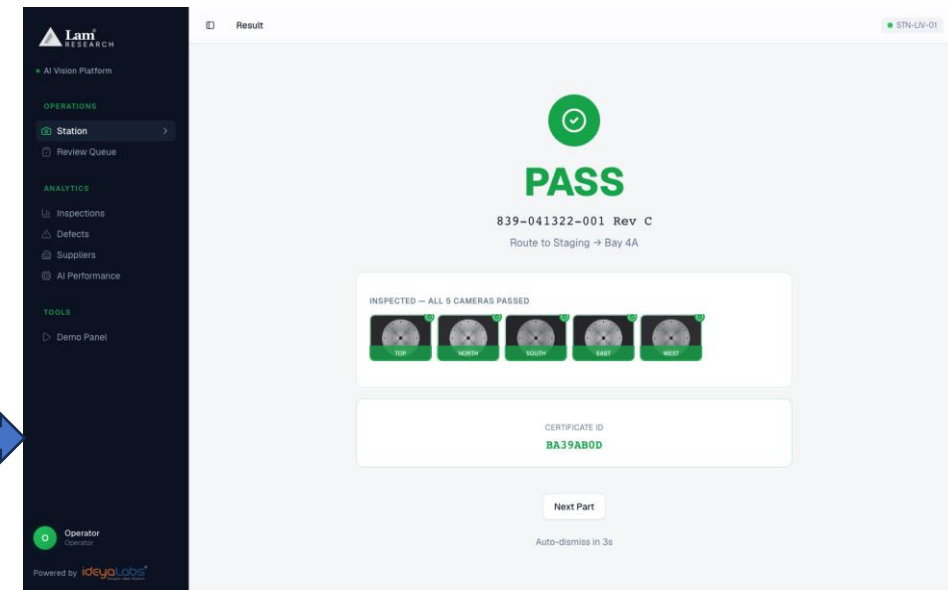


Step 5 – Multi-Angle Image Capture

Acquire high-resolution images from multiple angles for AI-powered inspection.

PASS Screen - Inspection Passed

Part meets all inspection criteria and proceeds to staging.

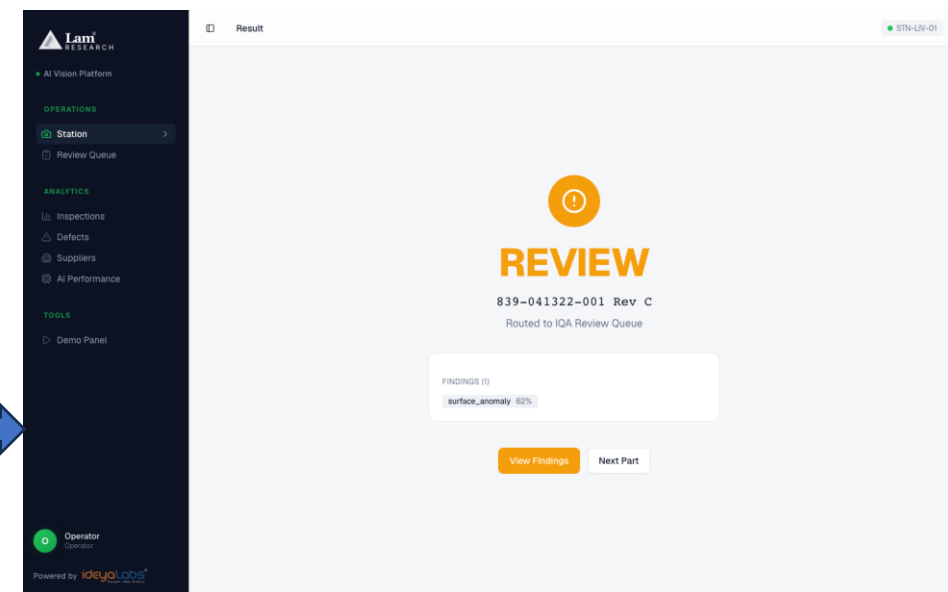


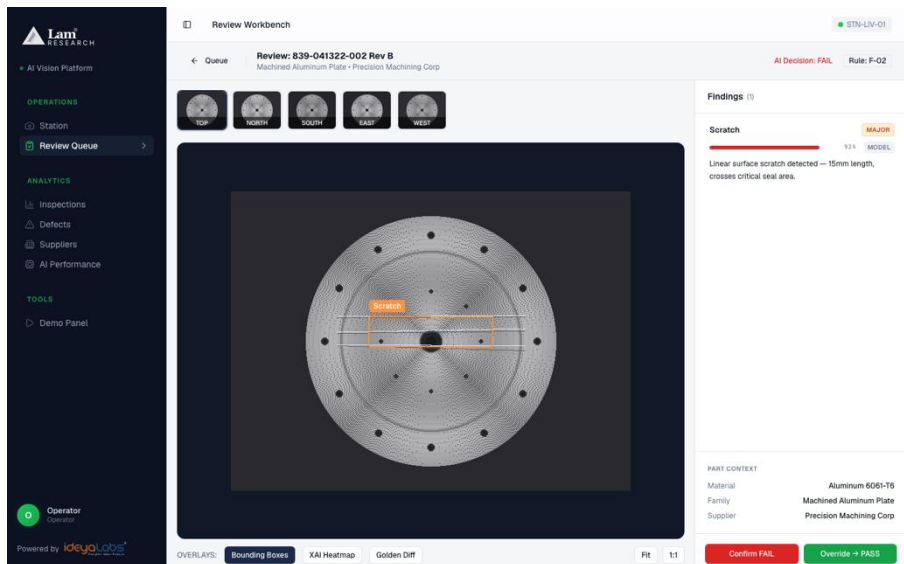
FAIL Screen - Defect Detected

Critical defects detected; part is routed to IQA for disposition.

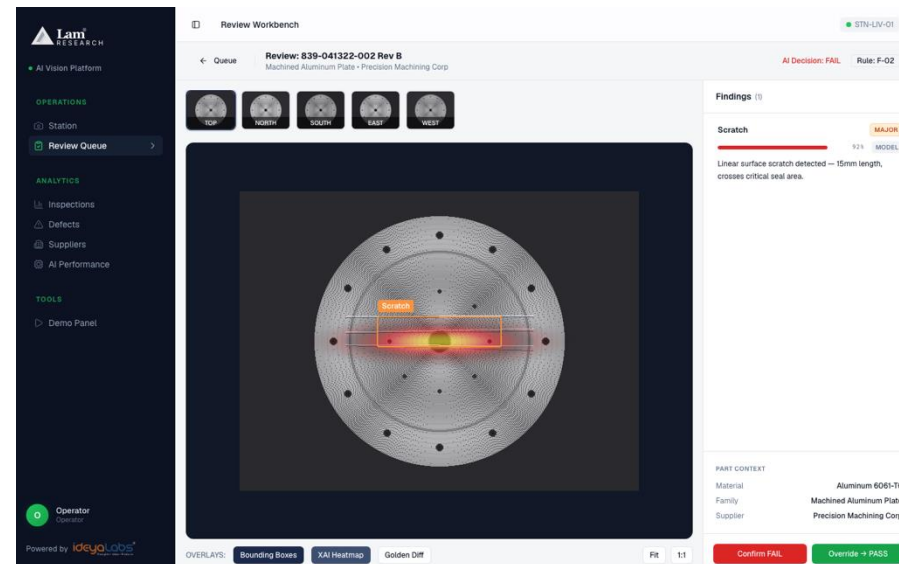
REVIEW Screen - Human Review Required

Low-confidence findings are escalated for quality engineer review.





Bounding Box - Defect Localization
Precisely identifies and highlights the detected scratch, enabling inspectors to quickly locate and verify the defect.



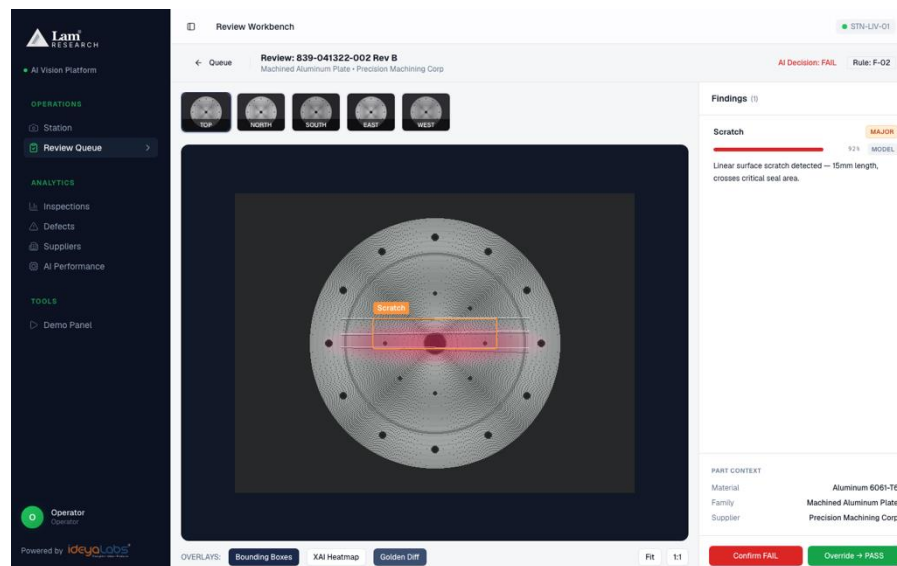
Bounding Box

XAI Heatmap

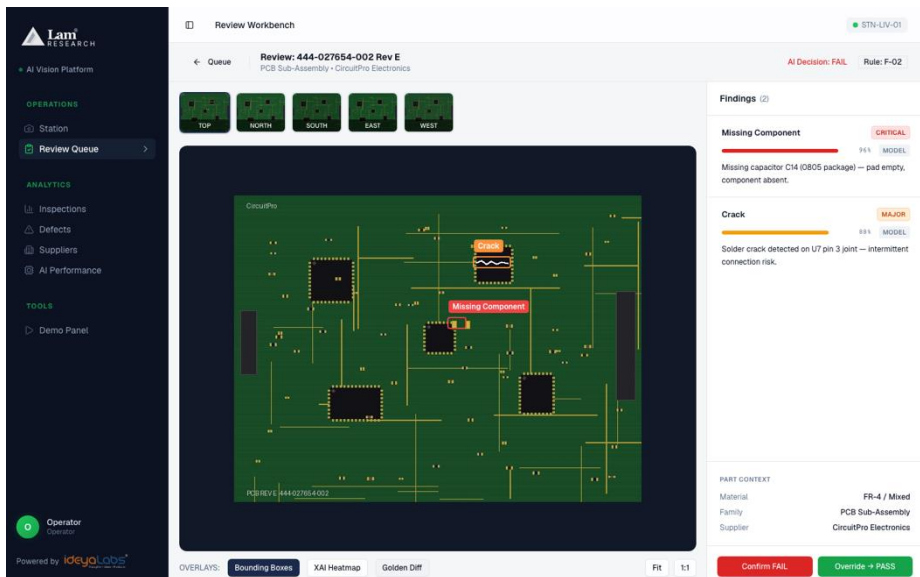
Golden Difference - Reference-Based Deviation Analysis

Compares the inspected part with the certified golden sample and highlights cosmetic deviations exceeding predefined tolerance limits.

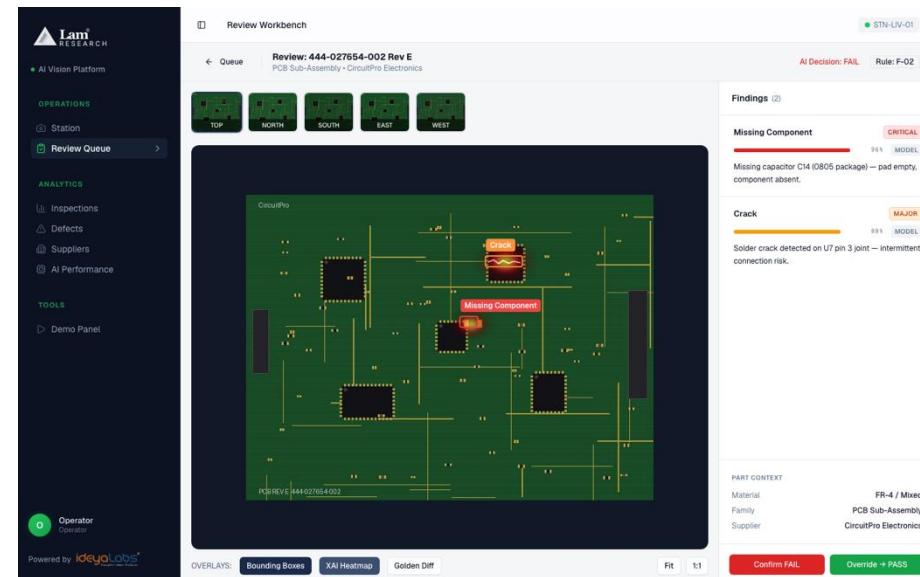
Golden Diff



XAI Heatmap- AI Decision Explainability
Visualizes the image regions that most influenced the AI's decision, providing transparency and confidence in the inspection results.



Bouding Box - Defect Localization
Highlights the exact location of the missing component and crack.



Bouding Box

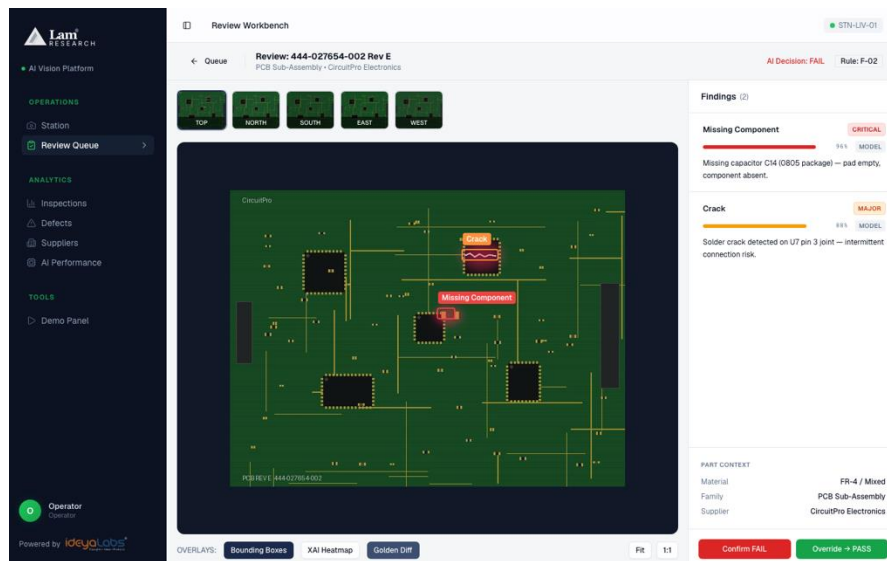
XAI Heatmap

Golden Difference - Reference-Based Deviation Analysis

Highlights deviations from the certified golden reference PCB.



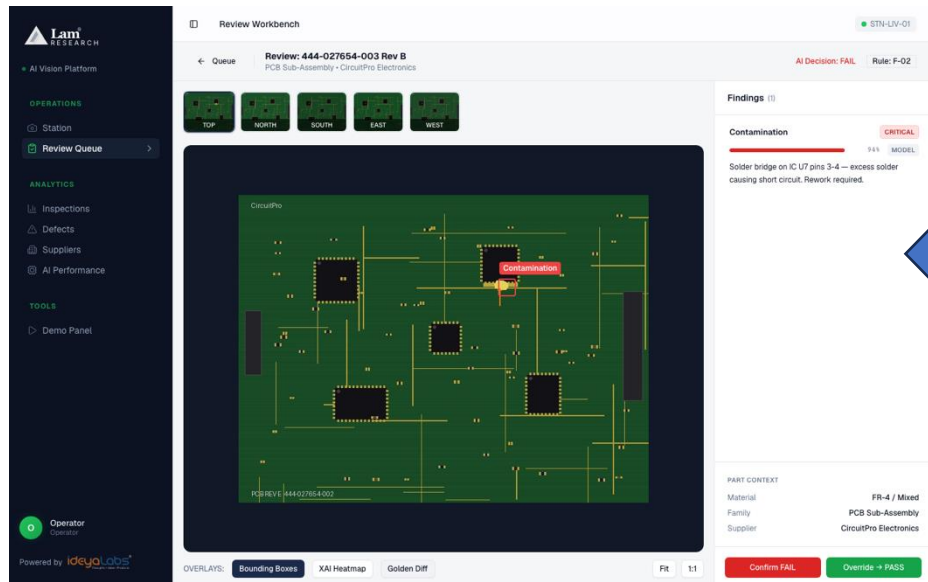
Golden Diff



XAI Heatmap- AI Decision Explainability

Shows the regions that influenced the AI's defect detection.

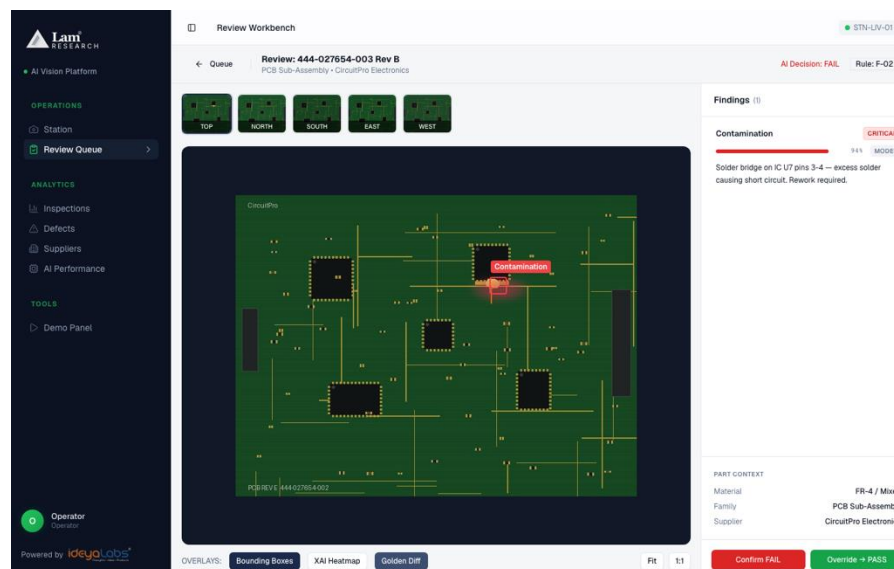




Bounding Box

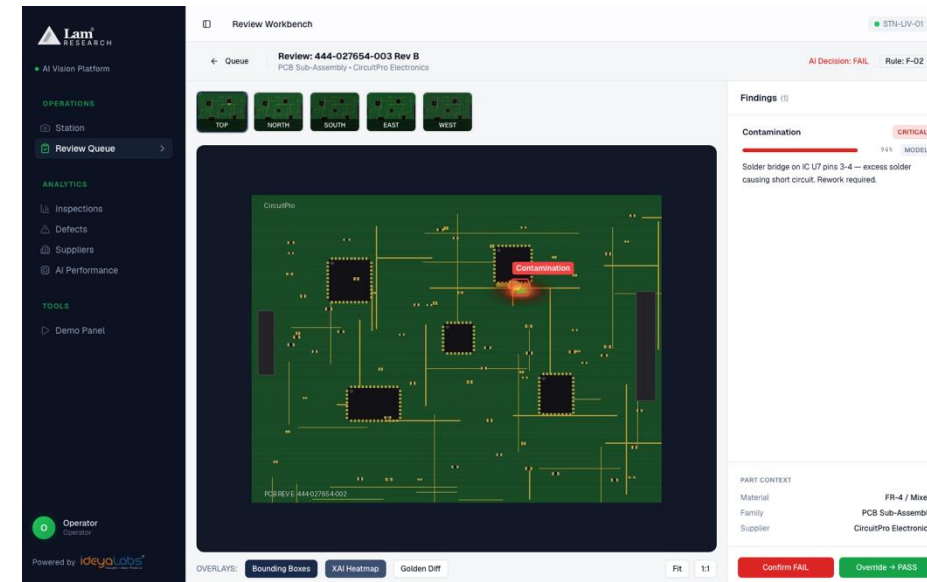
Golden Difference - Reference-Based Deviation Analysis
Highlights contamination and surface deviations from the certified golden sample.

Golden Diff



Bounding Box - Defect Localization

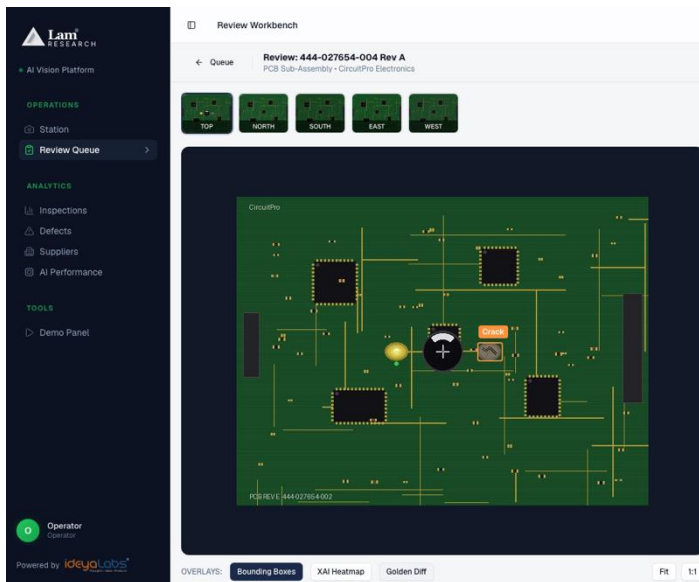
Highlights the exact location of the detected contamination



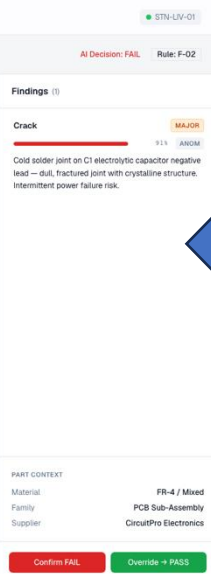
XAI Heatmap

XAI Heatmap- AI Decision Explainability
Shows the regions that influenced the AI's contamination detection.



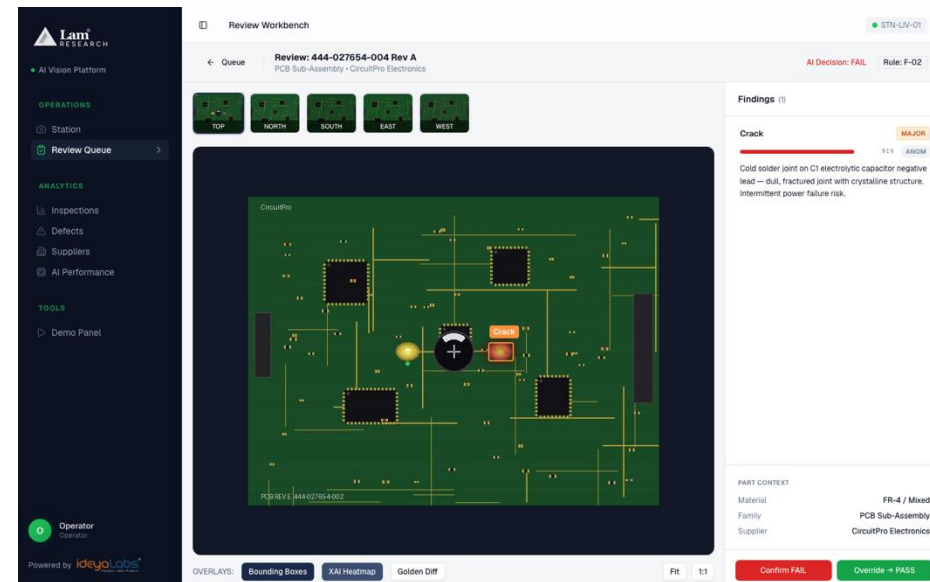


Bounding Box



Bounding Box - Defect Localization

Highlights the exact location of the detected crack



XAI Heatmap



XAI Heatmap- AI Decision Explainability

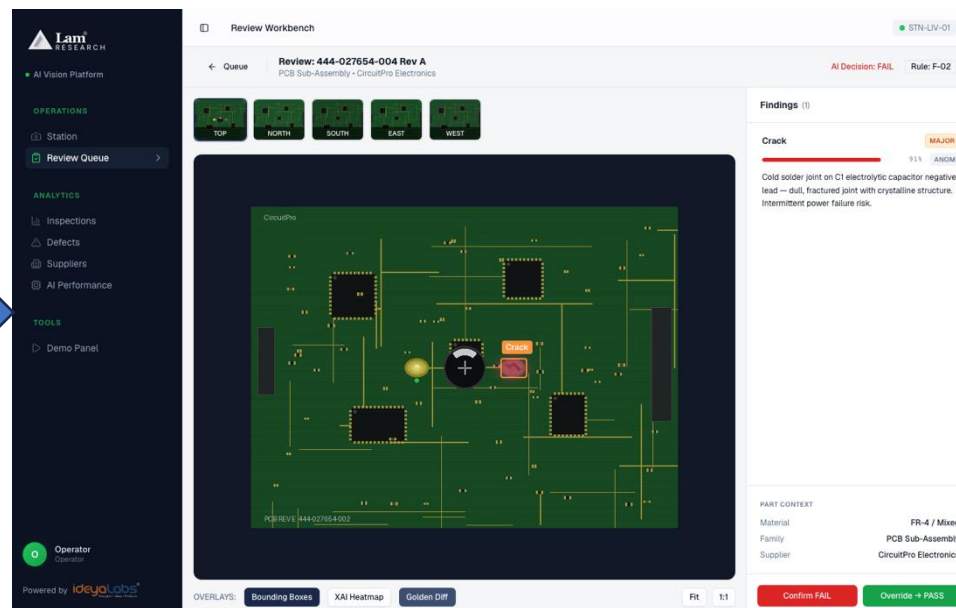
Shows the regions that influenced the AI's crack detection.

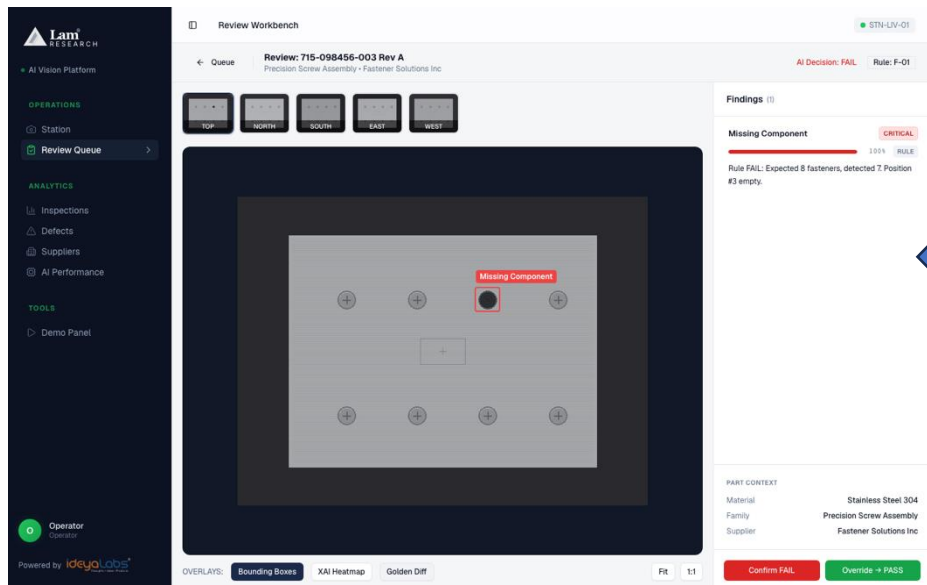
Golden Difference - Reference-Based Deviation Analysis

Highlights crack-related deviations from the certified golden sample.

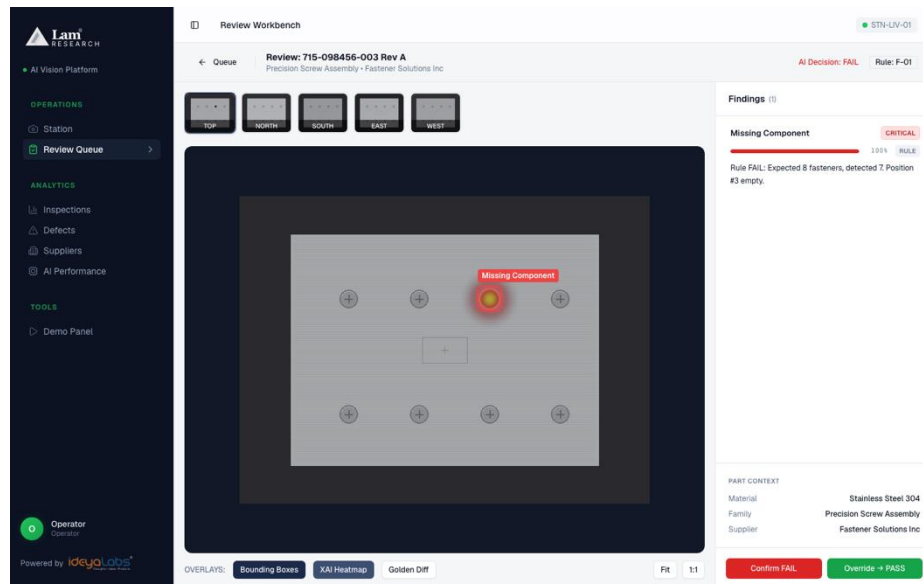


Golden Diff





Bounding Box - Defect Localization
Highlights the exact location of the missing component



XAI Heatmap

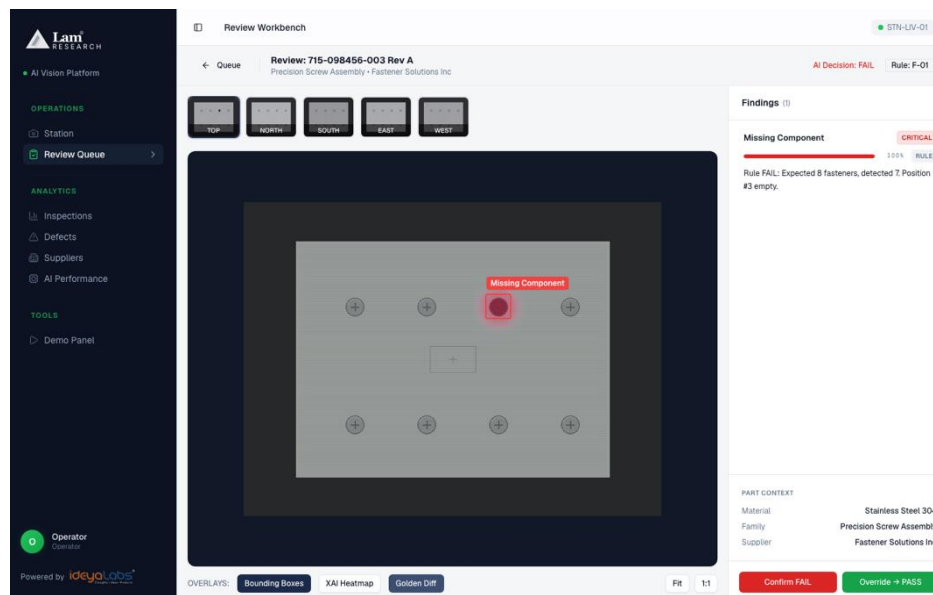


Golden Difference - Reference-Based Deviation Analysis

Highlights missing components by comparing against the certified golden sample.

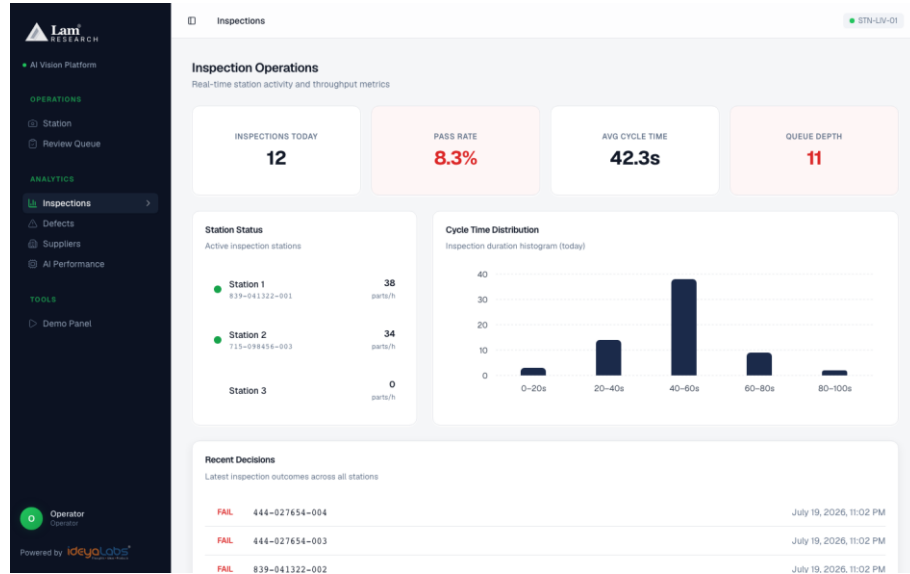


Golden Diff



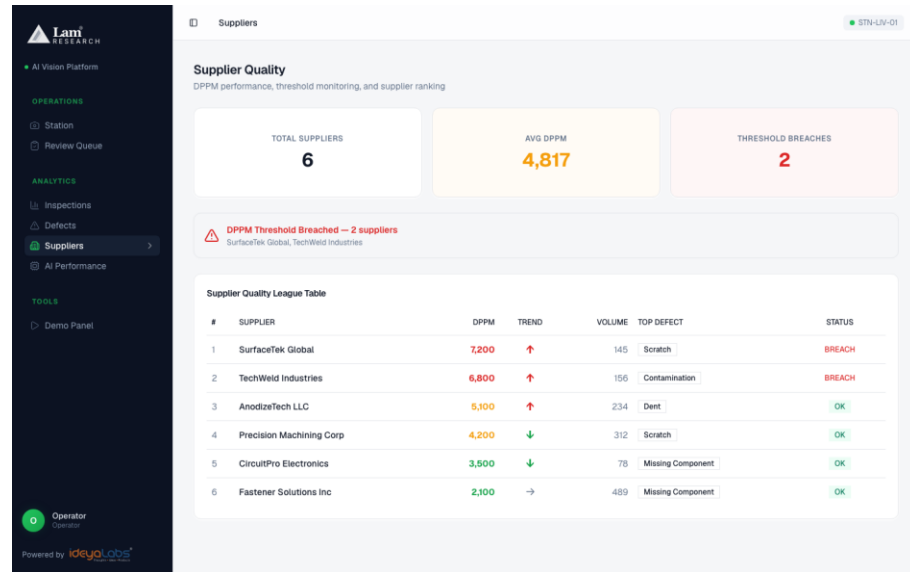
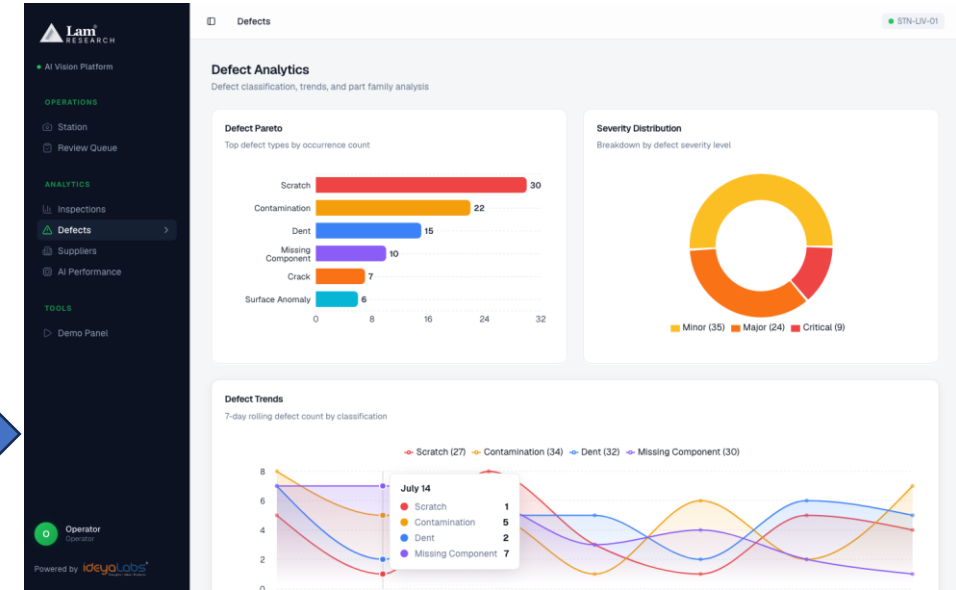
XAI Heatmap- AI Decision Explainability

Shows the regions that influenced the AI's missing component detection.



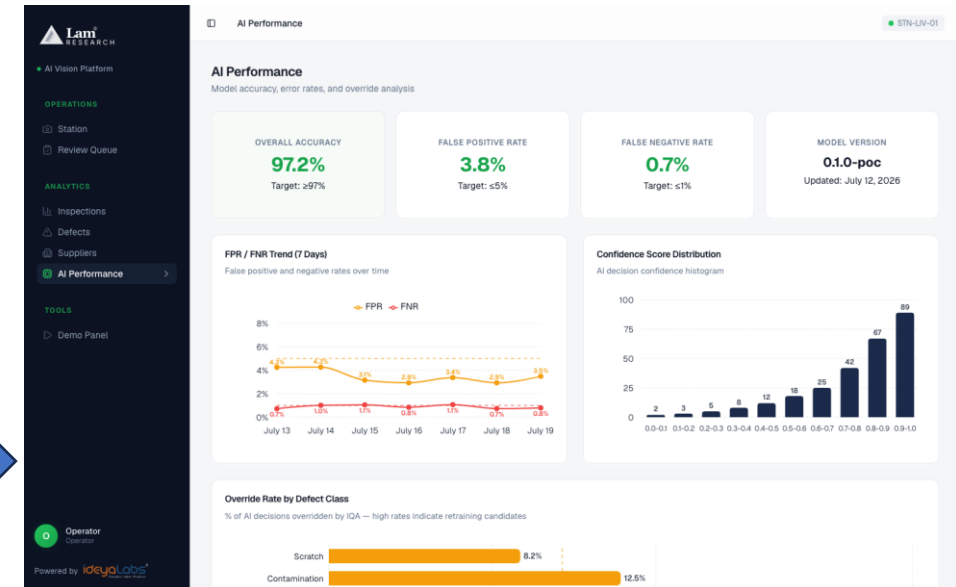
Inspection Operations Dashboard
Monitor real-time inspection throughput, cycle time, and station performance.

Defect Analytics Dashboard
Analyze defect trends, severity, and quality insights.



Supplier Quality Dashboard
Track supplier DPPM, quality trends, and performance rankings.

AI Performance Dashboard
Monitor AI accuracy, confidence, and model performance metrics.





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